Spectral Mixture Model Inspired Network Architectures for Hyperspectral Unmixing

Yuntao Qian, *Member, IEEE*, Fengchao Xiong, Qipeng Qian, and Jun Zhou, *Senior Member, IEEE*

***Abstract*— In many statistical hyperspectral unmixing approaches, the unmixing task is essentially an optimization problem given a defined linear or nonlinear spectral mixture model. However, most of the model inference algorithms require a time-consuming iterative procedure. On the other hand, neural networks have been recently used to estimate abundances given some training samples, or directly estimate endmembers and abundances simultaneously in an unsupervised setting. However, their disadvantages are clear: lack of interpretability and reliance on the large training set. Model-inspired neural networks are constructed by the problem model and its corresponding inference algorithm. It incorporates the prior knowledge of physical model and algorithm into network architecture, combining the advantages of model-based and learning-based methods. This article deeply unfolds the linear mixture model and the corresponding iterative shrinkage-thresholding algorithm (ISTA) to build two unmixing network architectures. The first assumes that the set of endmembers are known, and the deep unfolded ISTA model is only for abundance estimation; and the second is used for blind unmixing to estimate both endmembers and abundances at the same time. The networks can be trained by supervised and unsupervised schemes, respectively, with a small-size training set, and then, unmixing becomes a feedforward process, which is very fast since no iteration is required. The experimental results show their competitive performance compared with the state-of-the-art unmixing approaches.**

***Index Terms*— Deep unfolding, iterative methods, model-inspired network, spectral unmixing.**

I. Introduction

HYPERSPECTRAL imaging has been widely adopted in remote sensing, thanks to the spectral information contained in hundreds of continuous narrow bands. Due to the limited spatial resolution of imaging sensors, many pixels in a hyperspectral image (HSI) contain reflections from more than one type of materials in their receptive fields. Spectral unmixing aims to decompose a mixed pixel into a collection of constituent spectra, or endmembers, and their corresponding fractional abundances. This has become an important process to improve the utility of HSIs.

The estimation of endmembers and abundance can be done separately in a sequential pipeline [1]. It first determines the endmembers in an HSI using an endmember identification method, e.g., vertex component analysis (VCA) [2], or directly obtains all possible endmembers from a spectral library [3]. Then, the corresponding abundances can be considered as a regression problem with various constraints related to unmixing such as nonnegativity and sparsity. To solve the constrained regression problems, a number of iterative optimization algorithms have been proposed, which not only allow easy incorporation of constraints but also allow better regularization for an ill-conditioned model. A fully constrained least-squares (FCLS) algorithm was proposed for linear unmixing, which imposes both nonnegative and sum-to-one constraints on abundances [4]. Sparsity constraint is based on the fact that a mixed pixel always only contains a few constituted endmembers, especially if the spectral library contains a very large number of spectral signatures. Many sparsity-constrained linear regression algorithms can be used for abundance estimation [3], such as matching pursuit (MP), orthogonal matching pursuit (OMP), and iterative shrinkage thresholding algorithm (ISTA) [5], [6].

When both endmembers and abundance are estimated simultaneously, it is called blind unmixing. Nonnegative matrix factorization (NMF) is a widely used tool for blind unmixing, thanks to its inherent advantages in learning a part-based representation of data [7]. A number of variations of NMF have been proposed, most of which embed a range of constraints into the abundance and/or endmember estimation to incorporate spatial, spectral, and joint spectral–spatial information into NMF model [8]–[11]. In order to address information loss when converting a 3-D HSI into a 2-D matrix, nonnegative tensor factorization (NTF) was proposed for blind unmixing [12], [13]. The solutions for NMF and NTF are mainly based on iterative optimization algorithms, such as Lagrange multipliers, alternating direction method of multipliers [14], and adaptive half-thresholding algorithm [15]. In general, the main difficulties associated with these algorithms are how close the convergence is to the right solution and the speed of convergence.

Manuscript received October 27, 2019; revised December 30, 2019 and February 16, 2020; accepted March 18, 2020. Date of publication April 2, 2020; date of current version September 25, 2020. This work was supported in part by the 2030 National Key AI Program of China under Grant 2018AAA0100500, in part by the National Key Research and Development Program of China under Grant 2018YFB0505000, and in part by the National Natural Science Foundation of China under Grant 61571393 and Grant 61672240. *(Corresponding author: Yuntao Qian.)*

Yuntao Qian and Fengchao Xiong are with the Institute of Artificial Intelligence, College of Computer Science, Zhejiang University, Hangzhou 310027, China (e-mail: ytqian@zju.edu.cn; fcxiong@zju.edu.cn).

Qipeng Qian is with the School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: qipeng_qian@sjtu.edu.cn).

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia (e-mail: jun.zhou@griffith.edu.au).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.2982490

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

The above conventional unmixing approaches are dependent on the spectral mixture model and the associated model inference algorithms. During the past several years, there have been strong interests in introducing neural networks (NNs), especially deep neural networks (DNNs), for HSI processing [16]–[19]. NN-based unmixing methods can be categorized into supervised and unsupervised, which correspond to abundance estimation and blind unmixing, respectively. For supervised setting, the NN is trained with the spectra paired with their corresponding abundances at pixels in order to produce a function to predict the abundances [20]. The abundances of training samples can be obtained by using other abundance estimation methods or site-investigation/analysis. NNs and DNNs have a powerful ability to learn complex functions from supervised data, while these functions are not easy to deduce by conventional methods. For NN-based blind unmixing, the encoder–decoder architecture is adopted to obtain the endmember signatures and fractional abundances simultaneously [21]. It does not require any supervised information, overcoming the problem of leveraging supervised data of NN-based abundance estimation methods. In terms of representation learning theory, the blind unmixing procedure can be explained as finding a set of low-dimensional representations (i.e., abundances) that reconstruct the data with their corresponding bases (i.e., endmembers).

A significant disadvantage of NNs is network topology selection. The design of a sound network architecture normally requires a large number of trials and tuning [22]. Another well-known disadvantage is that NNs are closer to a black-box mechanism rather than problem-level formulations, so it is difficult to incorporate prior knowledge about the problem in hand. These disadvantages also cause the generalization and accuracy of learned functions to heavily depend on the quantity and quality of training samples.

In contrast to NNs, traditional model-based methods have explicit and physical significance. In an explicitly constructed model, the structure and form of latent variables and their relations link to specific semantics. Unfortunately, defining a good model to characterize these relationships is difficult in complex perceptual domains such as computer vision. For example, the spectral mixture models have clear physical interpretation in the Bayesian framework, but these models are the simplification of a real reflection interaction mechanism and environment, and are often difficult for the inference algorithms to obtain optimal solutions. On the other hand, NN-based methods impose very few assumptions on the physical model, acting as flexible function approximator learned from data rather than specifying the function by hand. NNs bypass the difficulties of accurately modeling physical mechanisms, but they do not always capture the nature of real applications.

Recently proposed model-inspired NNs combine and leverage these two trends. The deep unfolding technique is used for transferring a model and its associated iterative inference algorithm to an NN architecture [23], [24]. Specially, it unfolds the iterations into a layerwise structure to obtain novel NN-like architectures that can easily be optimized for learning an inference function. The model-inspired NNs have several distinctive properties: 1) a given level of accuracy can be reached with very fewer layers; 2) the architecture has a quite clear interpretation; 3) the requirement of accurate modeling is avoided; and 4) it is easy to incorporate domain knowledge into network architecture.

In this article, we propose two kinds of NNs, respectively, for abundance estimation and blind unmixing, both are inspired by linear mixture model (LMM) and ISTA. LMM is widely adopted in unmixing algorithms due to its simplicity and applicability. ISTA is a well-established optimization technique for solving linear inverse problems, which is adequate for solving large-scale problems, but is also known for quite slow convergency [25]. The proposed novel NN architectures for spectral unmixing bridge the gap between model-based and NN-based unmixing techniques, by making use of their respective advantages. We proposed the basic concept of model-inspired NN for abundance estimation in [26]. In comparison to our initial work, the novel contribution of this article includes the following.

1) We develop two novel network models inspired by LMM and ISTA for spectral unmixing, in which one is for abundance estimation, and the other is for blind unmixing. These two networks enjoy the advantages of both model-based and learning-based unmixing techniques, leading to well-defined interpretability, strong learning power, and little computational cost.
2) The details of network architectures are discussed, and the network learning algorithms are derived by gradient descent and backpropagation. As a general and efficient unmixing technique, it actually enables unfolding a spectral mixture model and the associated optimization algorithm into deep network architecture, which can be potentially applied to other unmixing models, such as nonlinear mixture model and spectral–spatial joint unmixing model.
3) The experimental results on both synthetic and real-world data show the competitive performance compared with the state-of-the-art unmixing approaches.

The rest of this article is organized as follows. Section II introduces some related works. In Section III, network architectures inspired by LMM and ISTA are given for abundance estimation and blind unmixing, respectively, and their learning algorithms and some implementation issues are also described. In Section IV, the experiments on both synthetic and real-world data sets are done to demonstrate the effectiveness of our methods. The conclusion and future work are summarized in Section V.

II. Related Work

In this section, we introduce some fundamental concepts and techniques that are related to the proposed methods.

A. LMM

LMM is based on the premise that the spectral signature of a mixed pixel is generated by a linear combination of endmembers' spectra. Given an HSI with p wavelength-indexed bands, LMM represents the spectrum of a pixel based upon R

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

endmembers and their corresponding abundances

$$\mathbf{x} = \mathbf{M}\mathbf{a} + \mathbf{e} \tag{1}$$

where $\mathbf{x}$ is a $p \times 1$ vector of the observed pixel spectrum, $\mathbf{M}$ is a $p \times R$ endmember matrix whose each column corresponds to an endmember spectrum, $\mathbf{a}$ is an $R \times 1$ vector of abundances for each endmember, and $\mathbf{e}$ is a $p \times 1$ vector of an additive noise representing the measurement errors.

Under the matrix notation, (1) can be denoted by

$$\mathbf{X} = \mathbf{M}\mathbf{A} + \mathbf{E} \tag{2}$$

where $\mathbf{X} \in \mathbb{R}^{p \times N}$ is an HSI that contains N pixels, $\mathbf{A} \in \mathbb{R}^{R \times N}$ is an abundance matrix, with each column representing the fractions for all endmembers at each pixel, and $\mathbf{E} \in \mathbb{R}^{p \times N}$ is the additive noise matrix.

In order to satisfy the physical interpretation, two constraints are usually imposed on (1), i.e., abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC). ANC indicates that the contribution of each endmember should be equal to or greater than zero, and ASC enforces that the total contributions of all endmembers sum to one. Moreover, the sparsity constraint is always added to LMM due to the fact that most of the mixed pixels consist of only a few endmembers.

B. Abundance Estimation as a Sparse Regression Problem

Assuming that the set of endmembers $\mathbf{M}$ have been extracted, the abundance estimation with sparsity and nonnegativity constraints can be defined as a sparse linear regression problem

$$\min_{\mathbf{a}} \frac{1}{2}\|\mathbf{x} - \mathbf{M}\mathbf{a}\|_2^2 + \lambda\|\mathbf{a}\|_1 \quad \texttt{s.t.} \quad \mathbf{a} \geq \mathbf{0} \tag{3}$$

where λ controls the strength of sparsity. This can be solved by some efficient algorithms [27], [28]. Among them, ISTA provides a solution using an iterative sequence of subproblems, each of which can be solved efficiently [5], [6], [25]. It iteratively calculates the following recursive equation to converge:

$$\mathbf{a}^{(k+1)} = h_\theta\left(\mathbf{W}\mathbf{x} + \mathbf{S}\mathbf{a}^{(k)}\right) \tag{4}$$

$\mathbf{W}$ and $\mathbf{S}$ are based on the endmember matrix $\mathbf{M}$

$$\mathbf{W} = \frac{1}{\alpha}\mathbf{M}^T, \quad \mathbf{S} = \mathbf{I} - \frac{1}{\alpha}\mathbf{M}^T\mathbf{M} \tag{5}$$

where $(\cdot)^T$ denotes the transpose of a matrix and α is the step size. h_θ is a shrinkage function that satisfies the nonnegativity constraint, defined as a componentwise vector one-sided shrinkage function with a vector of thresholds θ

$$h_\theta(\mathbf{y}) = \max(\mathbf{y} - \theta, 0). \tag{6}$$

In a standard ISTA, all the thresholds are set to $\theta = \lambda/\alpha$.

C. Blind Unmixing as an NMF Problem

NMF is a widely used solution to blind hyperspectral unmixing, which decomposes the observed HSI matrix $\mathbf{X}$ to estimate the endmember matrix $\mathbf{M}$ and abundance matrix $\mathbf{A}$ simultaneously. It can be modeled as the following L_1 regularized optimization problem:

$$\min_{\mathbf{A},\mathbf{M}} \frac{1}{2}\|\mathbf{X} - \mathbf{M}\mathbf{A}\|_2^2 + \lambda\|\mathbf{A}\|_1 \quad \texttt{s.t.} \quad \mathbf{A} \geq \mathbf{0},\ \mathbf{M} \geq \mathbf{0}. \tag{7}$$

The multiplicative iterative method is widely adopted to solve this problem. At each iteration, $\mathbf{M}$ and $\mathbf{A}$ are updated by

$$\mathbf{M} \leftarrow \mathbf{M}. * \mathbf{X}\mathbf{A}^T./\mathbf{M}\mathbf{A}\mathbf{A}^T \tag{8}$$

$$\mathbf{A} \leftarrow \mathbf{A}. * \mathbf{M}^T\mathbf{X}./\left(\mathbf{M}^T\mathbf{M}\mathbf{A} + \lambda\right) \tag{9}$$

where $.*$ and $./$ denote elementwise multiplication and division, respectively. Other optimization algorithms can be used to solve NMF and its variants, but like multiplicative iterative algorithm, they are also iterative and require a large number of iterations to converge [15].

D. NN-Based Unmixing Methods

Leveraging supervised data, NNs have been an effective solution proposed for abundance estimation. An NN maps the spectrum of a pixel $\mathbf{x}$ to its corresponding abundance vector $\mathbf{a}$

$$\mathbf{a} = f(\mathbf{x}). \tag{10}$$

The mapping function f is learned by the labeled pixels with known fractional abundances. However, it is very hard and expensive to acquire the precise fractional abundances of mixed pixels for NN training by doing site-investigation/analysis. Alternatively, the acquiring of the abundances of training pixels is through other unmixing approaches, but the quality of training samples is dependent on the method used.

Licciardi and Frate [29] proposed an NN of two modules for abundance estimation. The first module reduces the dimension of the input spectral vector with an auto-associative NN, while the second module maps the reduced input vector to the abundances with multilayer perceptron (MLP)-based fuzzy classification. The end-to-end pixel-based convolutional NN (CNN) and cube-based CNN for unmixing were proposed by Zhang *et al.* [20]. CNN is used to extract the discriminative features due to its good feature learning capability. Then, the features are used as the input to an MLP architecture at the last layer of the network for a pixel-based fuzzy classification procedure. The universal function approximation and end-to-end architecture of NN avoid explicit defining and solving spectral mixture models, making abundance estimation straightforward and fast. However, due to the massive parameters and the complex network structure, NN methods may not perform well when only a few training samples are available or the data quality is not good.

Blind unmixing can be considered as an unsupervised learning problem, for which autoencoders have been used as a common NN architecture for blind unmixing. Recent applications of autoencoders have addressed several factors

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

for blind unmixing, e.g., sparsity [30], nonnegativity [30], nonlinearity [31], deep representation [32], [33], outliers, and noises [34].

Autoencoder based pixel spectra reconstruction can be formulated as follows:

$$\min_{\mathbf{V},\mathbf{V}'} \|\mathbf{x} - \mathbf{V}'\varphi(\mathbf{V}\mathbf{x})\|_2^2 \tag{11}$$

where $\mathbf{x}$ is the spectra of a pixel, $\mathbf{V}$ is the weights of encoder while $\mathbf{V}'$ is the weights of the tired decoder, and φ denotes the activation function. If we compare (11) to the LMM in (1), their structures are analog to each other, i.e., the hidden representation $\varphi(\mathbf{V}\mathbf{x})$ generated by encoder can be interpreted as the abundance vector of a pixel, and the weights of decoder $\mathbf{V}'$ can be considered as a matrix whose columns are the endmembers.

NNs are the data-driven approach, so they do not require a specific statistical model of the problem in hand. Nevertheless, data-driven NNs are inherently "black-box" models, and unlike explicit physical/statistical models, the architecture and parameters of an NN are not particularly interpretable, making the choice of network architecture a process of trial-and-error. Some efforts have been made to use some prior domain knowledge to guide NN design, for instance, ANC, ASC, and sparsity are enforced [30], [32], [33]. However, it is still a challenging problem of building connections between physical/statistical models and NNs.

E. Model-Inspired NNs

Model-based methods use prior knowledge and intuition to reason at the problem level in devising inference algorithms, while NN-based methods use data and universal approximator to solve the problem at the computational level in learning inference functions. Model-inspired NNs aim to obtain the advantages of both model-based and NN-based methods by deep unfolding technique. Given a specific task, a model-based approach that requires an iterative inference algorithm is first constructed based on the task background. Second, unfold the iterations into a layerwise structure and then untie the model parameters across layers to obtain NN-like architectures that can easily be trained discriminatively using gradient-based methods [23], [35]. Different from the traditional model-based approaches requiring accuracy modeling that is always a very difficult task, the model used for deep unfolding only needs a very rough and broad definition of the task because the model-based approach is only used to construct the NN architecture, and the NN-based solution can be learned by training data.

Assume a general formulation of a model, determined by parameters Θ that specify the relationships between the output of interest $\mathbf{y}$ and the observed variables $\mathbf{x}$. In most cases, the closed-form solution of the model does not exist, so numerical iterative methods are always used. Given an iterative algorithm, $\mathbf{y}$ can be iteratively optimized by the following update rule:

$$\mathbf{y}^{(k+1)} = f_\Theta\left(\mathbf{x}, \mathbf{y}^{(k)}\right) \tag{12}$$

where k is the iteration index. After the updating is converged after K iterations, the final $\mathbf{y}$ is obtained by $\mathbf{y}^{(K)}$. The model-inspired NN, rather than considering the iterations as an algorithm, unfolds iterations as a sequence of layers in an NN-like architecture, and the parameters Θ are untied across layers as the transformation and activation function in each layer. Different from the model-based methods, Θ will be learned as the parameters of NN through training samples, so that the learned Θ' is not the same as that in the original model. This learning ability can make the actually required number of layers L much less than the number of iterations K required in the iterative algorithm.

Recently, several model-inspired NNs have been developed and applied for audio and speech processing [36], [37]. They have shown competitive performance against the state-of-the-art LSTM networks [24]. When applied to image denoising, a novel end-to-end deep network architecture based on a Gaussian conditional random field model [38] explicitly models the input noise variance and, hence, is capable of handling a range of noise levels. Furthermore, model-inspired NNs have shown to be promising in feature selection [39], sparse linear inverse problem [40], topic models [41], sparse coding [35], [42], and image reconstruction [43].

III. Unfolding LMM-Based ISTA Into Unmixing Networks

Deep unfolding uses the existing models and their inference algorithms as a starting point for NN architecture construction and initialization. In other words, it exploits model-based thinking to inspire, build, initialize, and improve NNs, which facilitates better interpretability, speeds up training convergence, and promotes better solutions. In this section, we use LMM and ISTA as the model and inference algorithm to build the unsupervised and supervised network architectures for abundance estimation and blind unmixing, respectively. We also derive their learning algorithms and discuss some implementation issues.

A. Supervised Model-Inspired NN for Abundance Estimation

We first discuss how to construct an LMM- and ISTA-inspired NN-like architecture that can be trained to directly infer the abundance vector of a mixed pixel. "Unfold" the iterations of ISTA in (4), thus resulting in a nonlinear, parameterized, feedforward architecture with a fixed depth, as shown in Fig. 1(a). In Fig. 1(a), the depth of the network is set as 3, each layer represents an iteration in ISTA, and the parameters of ISTA are untied across layers. This network can be trained to approximate the optimal abundance vector with a supervised data set $\{\mathbf{x}_n, \mathbf{a}_n\}_{n=1:N}$ with N samples. Its loss function is defined as follows:

$$\mathcal{L}_\Theta = \frac{1}{2N}\sum_{n=1}^{N} \|\hat{\mathbf{a}}_n - \mathbf{a}_n\|_2^2 = \frac{1}{2N}\sum_{n=1}^{N} \|f(\mathbf{x}_n, \Theta) - \mathbf{a}_n\|_2^2 \tag{13}$$

where $\hat{\mathbf{a}}$ is the model output, i.e., estimated abundance vector. $\Theta = \{\mathbf{W}', \mathbf{S}', \theta'\}$ is the parameter set of the network, which is originated from $\{\mathbf{W}, \mathbf{S}, \theta\}$ in (4) of ISTA. These parameters are learned during the training procedure in our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

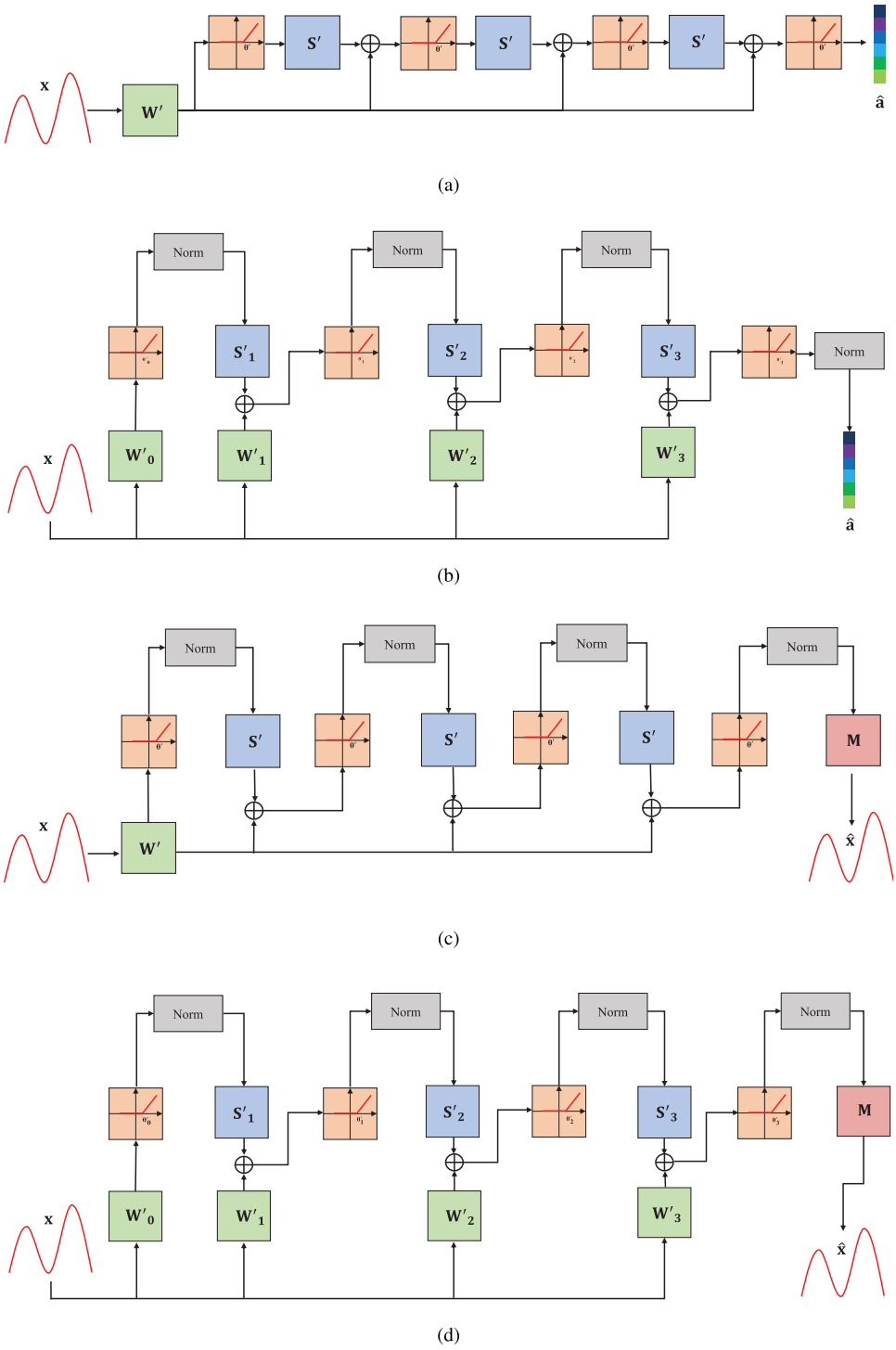


Fig. 1. Four NN architectures inspired from LMM and ISTA. (a) and (b) Abundance estimation. (c) and (d) Blind unmixing. (a) MNN-AE-1. (b) MNN-AE-2. (c) MNN-BU-1. (d) MNN-BU-2.

NN, but they are required to be known in the ISTA. The fixed parameters in ISTA are changed to be the learned parameters in model-inspired NN, so that the number of layers in our network is greatly less than the number of iterations required in the ISTA. In Fig. 1(a), the parameters are all shared in every layer. In order to increase the capacity of approximation and the flexibility of architecture, we can let $\Theta = \{\mathbf{W}', \mathbf{S}', \theta'\}$ unshared in each layer, i.e., $\Theta_l = \{\mathbf{W}'_l, \mathbf{S}'_l, \theta'_l\}_{l=1:L}$, where l is the layer index and L is the number of layers. This NN architecture is shown in Fig. 1(b). The total parameter set is $\Theta = \{\Theta_0, \ldots, \Theta_l, \ldots, \Theta_L\}$, in which $\Theta_0 = \{\mathbf{W}'_0, \theta'_0\}$. These two NNs are abbreviated as MNN-AE-1 and MNN-AE-2, respectively, in which MNN represents the model- and algorithm-inspired NN, AE represents the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

abundance estimation, and 1 and 2 represent the parameter shared and unshared across layers. Both MNN-AE-1 and MNN-AE-2 have their own merits, e.g., coupled architecture in MNN-AE-1 is closer to ISTA while decoupled one in MNN-AE-2 has better approximation when their network sizes are the same.

The training of MNN-AE-1 and MNN-AE-2 can be via the stochastic gradient descent algorithm [44] to minimize the loss function

$$\Theta^{(k+1)} = \Theta^{(k)} - \eta^{(k)} \frac{\partial \mathcal{L}(\Theta)}{d\Theta} \tag{14}$$

where $\eta^{(k)}$ is the learning rate. We can compute the derivatives of $\mathcal{L}$ with respect to each parameter recursively. Since the updating rules of MNN-AE-1 can be easily obtained from those of MNN-AE-2, here we only consider MNN-AE-2. With the help of backpropagation algorithm, we can calculate the gradients for all network parameters starting from the output down to the input in a sequence of $\Theta_L \to \cdots \to \Theta_l \to \cdots \to \Theta_0$ for each training iteration.

At first, we define some intermediate variables in MNN-AE-2 for the convenience of derivation

$$\mathbf{c}_l = \mathbf{W}_l \mathbf{x} + \mathbf{S}_l \hat{\mathbf{a}}_{l-1}, \quad l = 1 : L \tag{15}$$

$$\hat{\mathbf{a}}_l = \begin{cases} h_{\theta_l'}(\mathbf{c}_l), & l = 1 : L \\ h_{\theta_l'}(\mathbf{W}_0' \mathbf{x}), & l = 0. \end{cases} \tag{16}$$

Then, the gradient of loss with respect to parameters can be computed as follows:

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{a}}_L} = \hat{\mathbf{a}}_L - \mathbf{a} \tag{17}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{c}_l} = g_{\theta_l'}(\mathbf{c}_l) \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{a}}_l}, \quad l = 1 : L \tag{18}$$

$$\frac{\partial \mathcal{L}}{\partial \theta_l'} = -g_{\theta_l'}(\mathbf{c}_l) \frac{\partial \mathcal{L}}{\hat{\mathbf{a}}_l}, \quad l = 1 : L \tag{19}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_l'} = \frac{\partial \mathcal{L}}{\partial \mathbf{c}_l} \mathbf{x}^T, \quad l = 1 : L \tag{20}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{S}_l} = \frac{\partial \mathcal{L}}{\partial \mathbf{c}_l} \hat{\mathbf{a}}_{l-1}^T, \quad l = 1 : L \tag{21}$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{a}}_l} = \mathbf{S}_{l+1}^T \frac{\partial \mathcal{L}}{\partial \mathbf{c}_{l+1}}, \quad l = 1 : L - 1 \tag{22}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_0} = g_{\theta_0'}(\mathbf{W}_0' \mathbf{x}) \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{a}}_0} \mathbf{x}^T \tag{23}$$

$$\frac{\partial \mathcal{L}}{\partial \theta_0'} = -g_{\theta_0'}(\mathbf{W}_0' \mathbf{x}) \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{a}}_0} \tag{24}$$

where

$$g_\theta(\mathbf{y}) = \begin{cases} 1, & \mathbf{y} \geq \theta \\ 0, & \mathbf{y} < \theta. \end{cases} \tag{25}$$

As l starts from L and ends at 0, one iteration of backpropagation is finished.

The overall abundance estimation procedure includes the following steps.

1) Acquire a supervised data set $\{\mathbf{x}_n, \mathbf{a}_n\}_{n=1:N}$. Each sample contains the spectrum of a mixed pixel and its corresponding abundance vector, which can be obtained by any traditional unmixing methods.
2) Train MNN-AE-1 or MNN-AE-2 with stochastic gradient descent algorithm. Before learning, the model parameters are initialized with the prior known endmember set $\mathbf{M}$ and ISTA optimization parameter θ. Compared with conventional NNs, MNN-AE-1 and MNN-AE-2 are simpler, faster, and require less training samples.
3) Input any new mixed pixels, the learned MNN-AE-1 or MNN-AE-2 outputs its abundance vector. The speed of this step is very fast, thanks to that it is only a feedforward computation. It must be kept in mind that input samples should be drawn from the same distribution as the training samples, for example, all mixed pixels are collected by the same hyperspectral imager and on a similar geographical scene.

B. Unsupervised Model-Inspired NN for Blind Unmixing

Inspired by LMM and ISTA, we propose unsupervised NNs for blind unmixing, which simultaneously extract endmembers and estimate abundance. Fig. 1(c) and (d) shows two network architectures. The first has shared parameters in different layers as MNN-AE-1, while the second is alike MNN-AE-2 without shared parameters. They are called as MNN-BU-1 and MNN-BU-2, respectively, where BU represents the blind unmixing.

Compared Fig. 1(c) and (d) with Fig. 1(a) and (b), their main difference is that MNN-BU-1 and MNN-BU-2 add the endmember matrix $\mathbf{M}$ as the last layer, and then, their output is changed to be the reconstructed input $\hat{\mathbf{x}}$ instead of $\hat{\mathbf{a}}$ in MNN-AE-1 and MNN-AE-2. In addition, in order to make the abundance vector of a pixel satisfy the ASC constraint, we add normalization operators in the architecture. Obviously, if the endmember matrix $\mathbf{M}$ is prior known and fixed in MNN-BU-1 and MNN-BU-2, they are equal to MNN-AE-1 and MNN-AE-2, in which the abundance vector of a training sample $\mathbf{x}$ is calculated by solving the linear equation of LMM using the Moore–Penrose Inverse of $\mathbf{M}$, i.e., $\mathbf{a} = \mathbf{M}^+\mathbf{x}$. Consequently, once the endmembers are assumed to be known, the use of our unsupervised NN is not conceptually different from the using supervised MNN-AE-1 and MNN-AE-2. Their only difference is that in the supervised MNN-AE-1 and MNN-AE-2, the abundances of training samples can be obtained by various abundance estimation methods, while in MNN-BU-1 and MNN-BU-2, the abundances are hiddenly estimated by the Moore–Penrose inverse.

If the endmembers are not known, $\mathbf{M}$ in MNN-BU-1 and MNN-BU-2 can be set as the variables/parameters of NNs that must be learned in the training procedure, making the training problem completely unsupervised, i.e., not only do not require any labeled training pixels but also simultaneously estimate both endmembers and abundances. The unsupervised MNN-BU-1 and MNN-BU-2 can be considered as an encoder–decoder architecture. The part before the last layer is an encoder that transforms the pixel's original spectrum into an encoded vector (abundance vector). The last layer is a decoder that restores the pixel's spectrum from the encoded data. The unsupervised learning can be completed by minimizing loss function $\mathcal{L}$ with respect to the variables of both encoder and decoder, i.e., Θ and $\mathbf{M}$, where $\Theta = \{\mathbf{W}', \mathbf{S}', \theta'\}$ in MNN-BU-1

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

and $\Theta = \{\Theta_0, \ldots, \Theta_l, \ldots, \Theta_L\}$ with $\Theta_l = \{\mathbf{W}'_l, \mathbf{S}'_l, \theta'_l\}$ in MNN-BU-2. Here, we assume that these two sets of variables are untied, i.e., $\Theta = \{\mathbf{W}', \mathbf{S}', \theta'\}$ and $\mathbf{M}$ are not enforced by any correlated constraints. Based on the conclusion in [34], it is beneficial to blind unmixing if the encoder and decoder are independent.

The loss function is defined as follows:

$$\mathcal{L}_{\Theta,\mathbf{M}} = \frac{1}{2N}\sum_{n=1}^{N} \|\hat{\mathbf{x}}_n - \mathbf{x}_n\|_2^2 = \frac{1}{2N}\sum_{n=1}^{N} \|f(\mathbf{x}_n, \Theta, \mathbf{M}) - \mathbf{x}_n\|_2^2 \tag{26}$$

where $\hat{\mathbf{x}}$ is the model output, i.e., the reconstructed input pixel's spectrum.

Like the training of MNN-AE-1 and MNN-AE-2, the gradient of loss with respect to parameters of MNN-BU-1 and MNN-BU-2 can also be computed. The gradients of loss with respect to the last two variables are as follows:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{M}} = (\mathbf{M}\hat{\mathbf{a}}_L - \mathbf{x})\hat{\mathbf{a}}_L^T \tag{27}$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{a}}_L} = \mathbf{M}^T(\mathbf{M}\hat{\mathbf{a}}_L - \mathbf{x}). \tag{28}$$

The gradients for other variables are the same as those in MNN-AE-1 and MNN-AE-2. In order to make the estimated endmembers $\mathbf{M}$ are nonnegative, after each update step, we enforce their negative entities to a tiny value, i.e., e^{-6}.

As MNN-AE-1 and MNN-AE-2 are unsupervised NNs, there are two schemes of accomplishing blind unmixing.

1) Select some pixels as training samples for parameter learning of network. After the training procedure of MNN-BU-1 or MNN-BU-2 is finished, all variables/parameters of the network are determined. The learned $\mathbf{M}$ can be considered as the estimated endmembers. When the spectrum of a pixel is input into the trained network, $\hat{a_L}$, that is, the output item before $\mathbf{M}$, can be considered as the estimated abundances of this input pixel.
2) All pixels need to be unmixed are used as training samples, i.e., the training set and the testing set are the same in unsupervised learning. After the training procedure of MNN-BU-1 or MNN-BU-2 is finished, the learned $\mathbf{M}$ is also considered as the estimated endmembers, and $\hat{a_L}$ of a training pixel is taken as the abundances of this pixel.

IV. Experiments

In this section, we present the experimental results on both synthetic and real-world data to thoroughly evaluate the effectiveness of proposed unmixing methods. The spectral angle distance (SAD) and root-mean-square error (RMSE) are utilized to quantitatively evaluate the performance of unmixing. SAD is a criterion that measures the dissimilarity between the rth endmember $\mathbf{m}_r$ and its associated estimation $\hat{\mathbf{m}}_r$, which is defined as follows:

$$\mathrm{SAD}_r = \arccos\left(\frac{\mathbf{m}_r^T \hat{\mathbf{m}}_r}{\|\mathbf{m}_r\|\|\hat{\mathbf{m}}_r\|}\right). \tag{29}$$

RMSE is used to compare the error between the reference abundance map $\mathbf{A}_r$ of the rth endmember and its estimated map $\hat{\mathbf{A}}_r$, which is defined as follows:

$$\mathrm{RMSE}_r = \left(\frac{1}{N} \mid \mathbf{A}_r - \hat{\mathbf{A}}_r \mid^2\right)^{\frac{1}{2}} \tag{30}$$

where N is the number of pixels used for test. The average SADs and RMSEs of all the endmembers are calculated to compare the performance of unmixing methods.

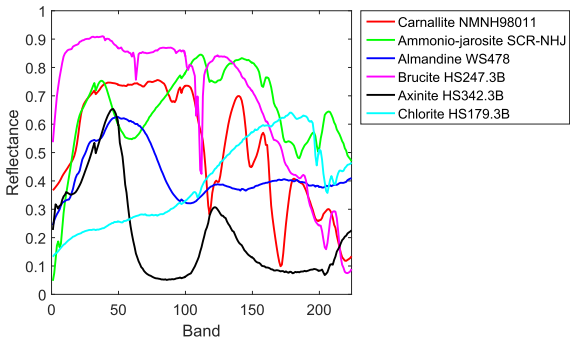


Fig. 2. Spectral signatures of six endmembers used in the synthetic data.

A. Synthetic Data

Similar to Jia and Qian [7], the synthetic data are produced by two steps: endmember generation and abundance generation. The endmembers are generated by selecting six pure signatures (Carnallite, Ammonio-jarosite, Almandine, Brucite, Axinite, and Chlonte) from the USGS library. They contain 224 bands with the wavelengths covering from 0.38 to 2.5 μm. Fig. 2 demonstrates the spectral signatures of these endmembers.

The abundances are generated by the following steps.

1) z^2 nonoverlapping patches with the same size of $z \times z$ pixels are extracted from a $z^2 \times z^2$ synthetic image.
2) Each pixel of a region is assigned with two kinds of randomly selected endmembers, whose fractions are initialized with β and $1 - \beta$, respectively.
3) Each abundance map is convolved with a Gaussian filter, whose variance is set as 2 to make the resulted map highly mixed.
4) The fractions of all the endmembers in each pixel are rescaled, by which the sum-to-one constraint of fractions is met.

In order to test the sensitivity of unmixing methods to noise, the clean HSI is corrupted by adding zero-mean white Gaussian noise with prespecified signal-to-noise ratio (SNR) defined as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{E\left[\mathbf{y}^T\mathbf{y}\right]}{E\left[\mathbf{e}^T\mathbf{e}\right]} \tag{31}$$

where $\mathbf{y}$ and $\mathbf{e}$ are the clean signal and the noise at a pixel and $E[\cdot]$ denotes the expectation operator. In all experiments on synthetic data, we set the number of endmembers as $R = 6$ and the patch size as $z = 8$, so the size of image is $64 \times 64 = 4096$ pixels.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

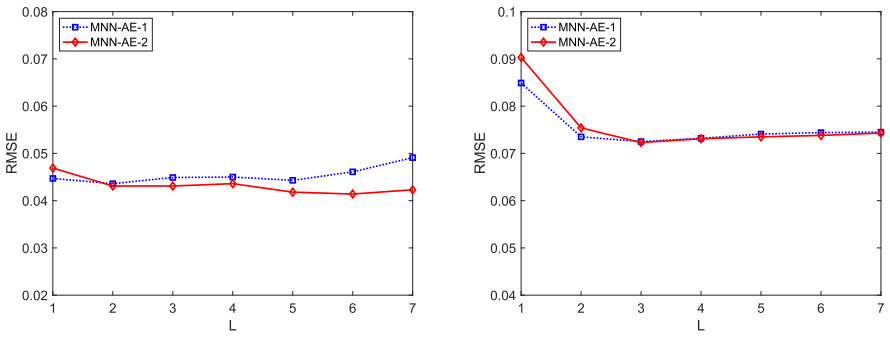


Fig. 3. RMSE of MNN-AE-1 and MNN-AE-2 with respect to different numbers of layers under two training conditions, respectively. (a) Ground truth. (b) VCA-FCLS.

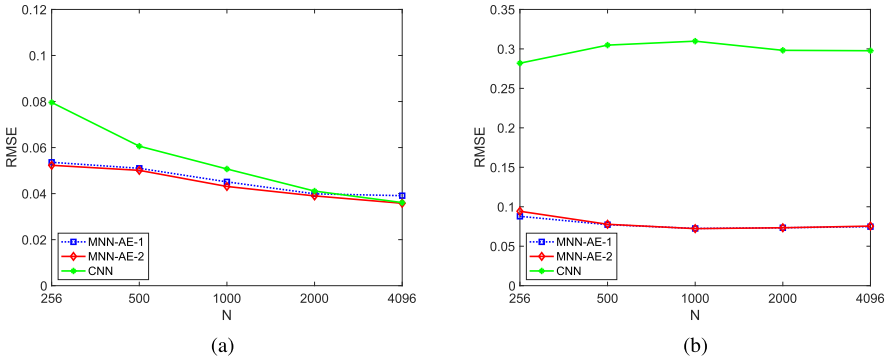


Fig. 4. RMSE of MNN-AE-1, MNN-AE-2, and CNN with respect to different numbers of training samples under two training conditions, respectively. (a) Ground truth. (b) VCA-FCLS.

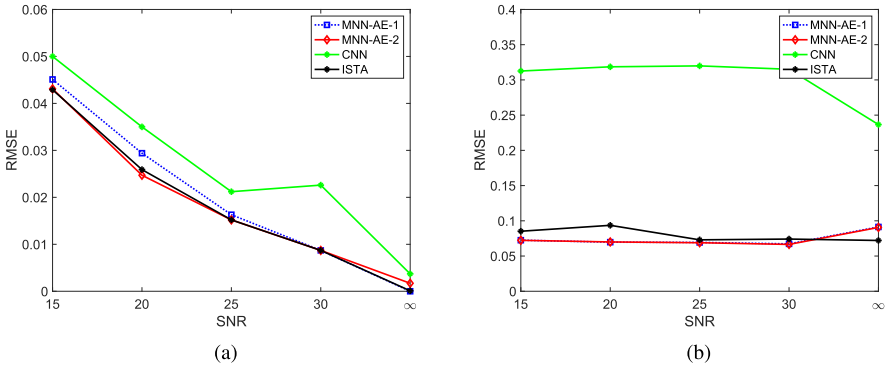


Fig. 5. RMSE of all four abundance estimation methods with respect to different noise levels under two training conditions, respectively. (a) Ground truth. (b) VCA-FCLS.

B. Abundance Estimation

Here, we conduct a series of experiments to illustrate the performances of MNN-AE-1 and MNN-AE-2 in abundance estimation. Pixel-based CNN [20] and ISTA [5] are used as alternative methods for comparison.

1) Abundance Estimation on Synthetic Data: At first, we evaluate the proposed methods on synthetic data. The training pixels of MNN-AE-1, MNN-AE-2, and CNN are randomly selected from a synthetic HSI, while the remaining pixels are used as test samples. The experiments are done under two settings. In the first setting, for MNN-AE-1, MNN-AE-2, and CNN, the abundances of the training samples come from the ground truth data, while for ISTA, the endmembers are also fully precise, and all of them can be obtained in synthetic generation procedure. This experiment evaluates the performance when the known information is completely right. For the

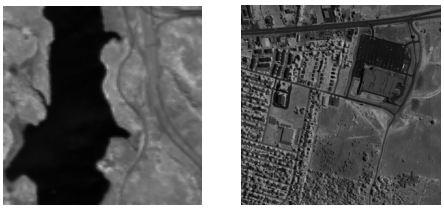


Fig. 6. Band images of two HSIs. (a) Jasper Ridge. (b) Urban.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

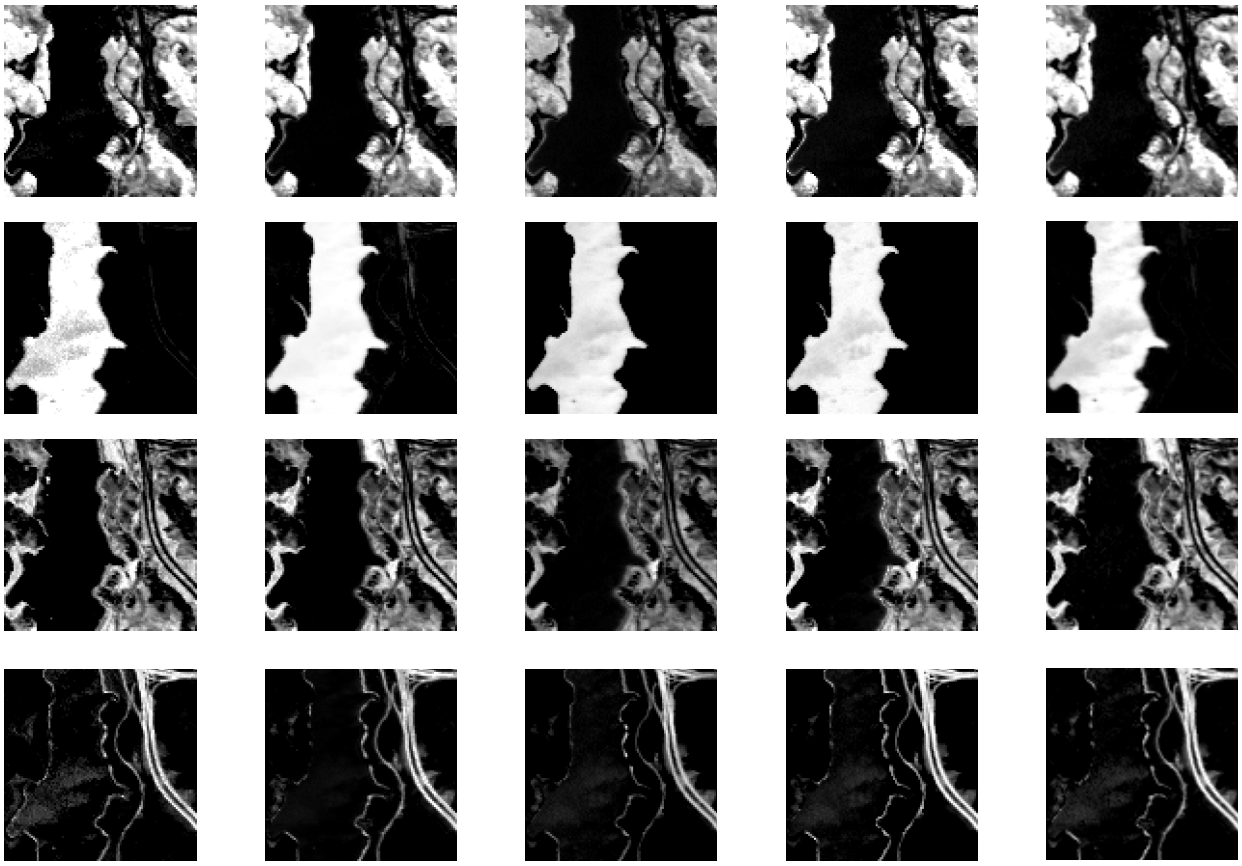


Fig. 7. Estimated abundance maps by four methods on the Jasper Ridge data set. (From top to bottom) Rows are the abundance maps of tree, water, soil, and road, respectively. (a) CNN. (b) ISTA. (c) MNN-AE-1. (d) MNN-AE-2. (e) Reference.

second setting, the above endmembers and abundances are obtained by VCA-FCLS, i.e., the endmembers are extracted by VCA [2], and then, the abundances are estimated by FCLS [4]. This experiment evaluates the performance when the known information is not perfect. This case is very common in real applications.

a) Impact of the number of layers: This experiment analyzes the impact of the number of network layers on the unmixing performance in our methods. In this experiment, the synthetic data are generated with $\beta = 0.8$, SNR $= 15$ dB, the number of training samples is set as 1000, and the numbers of layers is set from 1 to 7. Usually, a network with more layers leads to better approximation but introduces more parameters to be learned and may cause overfitting problem. Fig. 3 shows the results of MNN-AE-1 and MNN-AE-2 with two experimental settings, i.e., absolutely correct training samples and VCA-FCLS-based training samples. We can see that both of them need very few layers (2 or 3) to reach a favorable performance under two different settings, which demonstrate that our networks have strong learning power even with a very small network size. In the following experiments, we set $L = 2$ for our abundance estimation networks.

b) Impact of the number of training samples: This experiment analyzes the impact of the number of training samples on unmixing performance. The data in this experiment are also generated with $\beta = 0.8$, SNR $= 15$ dB. The architecture of pixel-based CNN is set as recommended in [20]. The sizes of training sets are set as $\{256, 500, 1000, 2000, 4096\}$. The results are given in Fig. 4, thus showing that the performance of our two networks is better than that of CNN, especially when the size of training samples is small and the quality of training samples is not perfect. This experiment demonstrates that our model-inspired NNs have distinct advantages compared with ordinary NNs.

c) Robustness to noise: This experiment evaluates the robustness of the abundance estimation methods to noise. The generated clean HSIs are polluted by different levels of noise with SNR $= \{15$ dB, 20 dB, 25 dB, 30 dB, $\infty\}$. The results are shown in Fig. 5, from which we can find that the performance of all methods becomes better as the noise level decreases when the ground truth data are used as training samples and known endmembers, but the noise level has no obvious impact on all four methods when the training samples or the known endmembers are not perfect. This may be because more training samples do not play a positive role when their own quality is not very good. ISTA has almost the same performance with our methods because our NNs are unfolded from ISTA. In particular, when there is no noise in the data (SNR $= \infty$), the performance of our method is not better than ISTA, as unfolding ISTA is only an approximation of original ISTA. This fact does not hold when the data are corrupted by noise, so that sometimes our method might have better performance than the original ISTA.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

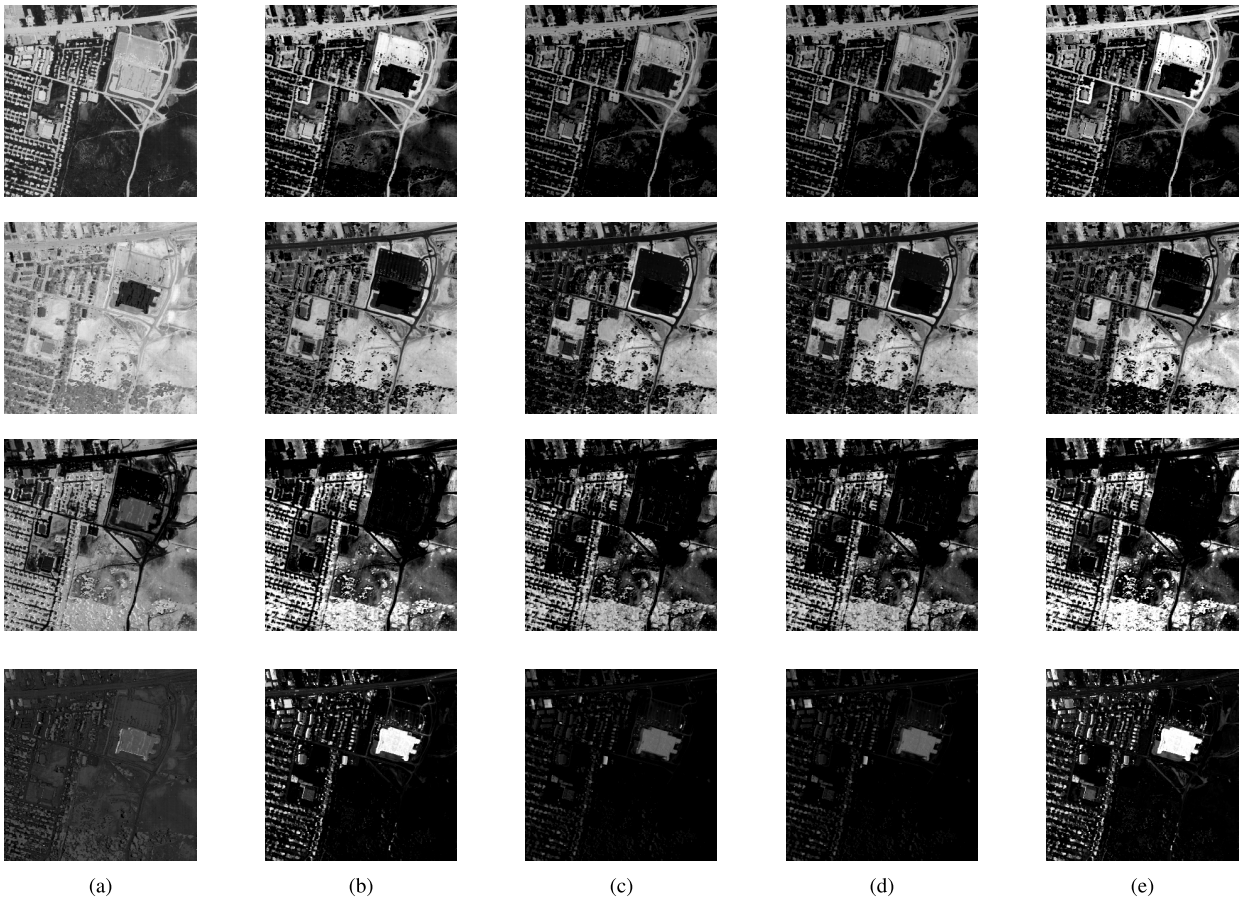


Fig. 8. Estimated abundance maps by four methods on the HYDICE Urban data set. (From top to bottom) Rows are the abundance maps of asphalt, grass, tree, and roof, respectively. (a) CNN. (b) ISTA. (c) MNN-AE-1. (d) MNN-AE-2. (e) Reference.

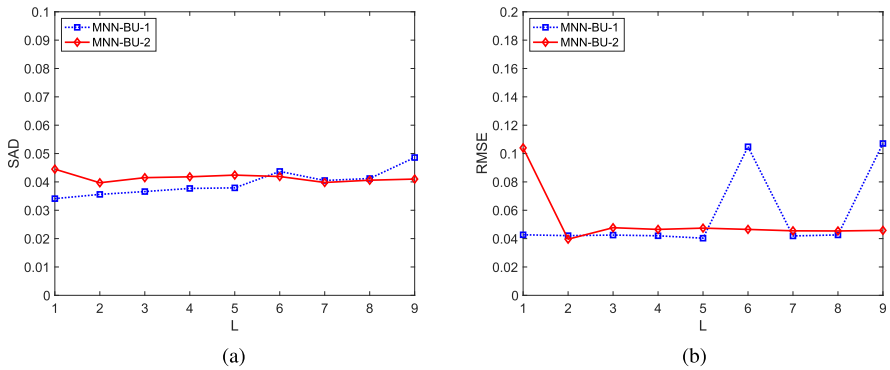


Fig. 9. Performance of MNN-BU-1 and MNN-BU-2 with respect to the numbers of layers in terms of (a) SAD and (b) RMSE.

d) Running time comparison: This experiment compares the running time of all the methods. The synthetic image is generated by setting $\beta = 0.8$, SNR $= 25$ dB. A Windows machine equipped with Intel Core i5-4440 CPU at 3.10 GHz and 16-GB RAM is used. All the algorithms are implemented using Python and MATLAB. For an HSI containing p bands and R endmembers, the number of parameters of MNN-AE-1 with L layers is $O(p + R + 1)R$, and MNN-AE-2 requires $O(p + R + 1)RL$ parameters. For CNN, the number of parameters is $O(3600p + 151R)$ according to its network structure [20], which is larger than those of MNN-AE-1 and MNN-AE-2.

In order to show the efficiency of the proposed method applied on a large data set, we add another synthetic HSI with a size of 640×640 using the same generating method. For both of two synthetic data sets with different sizes, the same number of training pixels, 1000, is used for MNN-AE-1, MNN-AE-2, and CNN, and all pixels are used for the test. Table I reports the running time of ISTA, CNN, MNN-AE-1, and MNN-AE-2 in two data sets. Compared with ISTA whose

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

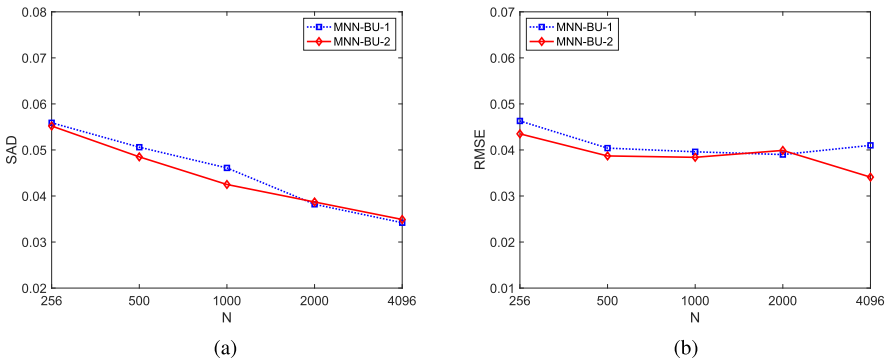


Fig. 10. Performance of MNN-BU-1 and MNN-BU-2 with respect to training samples in terms of (a) SAD and (b) RMSE.

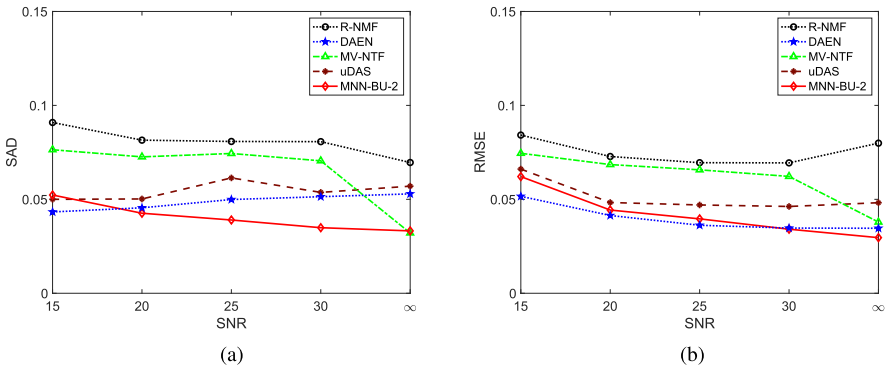


Fig. 11. Performance of all five blind unmixing methods with respect to different noise levels in terms of (a) SAD and (b) RMSE.

TABLE I
RUNNING TIME COMPARISON ON TWO SYNTHETIC DATA SETS WITH DIFFERENT SIZES

Time (s)	ISTA	CNN	MNN-AE-1	MNN-AE-2
64×64	878.2	3065.0	5.4	7.5
640×640	89839	3115.3	6.8	8.2

running time is approximately linear with the size of data set, three learning-based methods are almost unaffected by the size of data set, i.e., they are of well-scalability. This is because the training time of MNN-AE-1, MNN-AE-2, and CNN is dependent on the size of training set, and after training step, the feedforward computation of a learned network used for estimating the abundance vector of a mixed pixel is very fast. Also as a learning-based method, CNN takes dramatically more time to train the network, thus demonstrating that our network architectures have a smaller size and better convergence performance.

2) Abundance Estimation on Real-World Data: Now, we turn to the experiments on two widely used real data sets, Jasper Ridge and HYDICE Urban. Their reference endmembers and abundances are obtained by the method of Qian *et al.* [45]. First, several highly likely pure pixels are selected from an HSI, then, their average signature is considered as a reference endmember, and finally, its corresponding abundance map is obtained by using CVX MATLAB software after all the endmembers are ready. As these two data sets have some nearly pure pixels, it is generally believed that the reference endmembers obtained by pure pixel-based endmember extraction method are very close to the true endmembers. On the contrary, CVX MATLAB is only considered as a specific abundance estimation method, which can get an average-level abundance estimation by solving LMM-based convex optimization problem, so the reference abundances obtained by CVX MATLAB are used as the approximate solutions rather than the true or optimal abundances. Due to this reason, the quantitative evaluation of abundances (RMSE) is not adopted in these two real-world data sets for performance comparison, but rather only the visual comparison is provided.

a) Jasper Ridge Data Set: Jasper Ridge data set was acquired by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Jasper Ridge in central California, USA. The original image contains 512×614 pixels and 224 electromagnetic bands covering the electromagnetic spectrum from 380 to 2500 nm with a resolution of nearly 10 nm. Bands (1–3, 108–112, 154–166, and 220–224) are removed because of being affected by noises and water-vapor absorption, thus yielding 198 bands. A subscene of 100×100 pixels is cropped for the experiment, whose 80th band image is shown in Fig. 6(a). It is assumed that there are four types of endmembers, i.e., soil, water, tree, and road. We randomly select 1000 pixels as training samples for MNN-AE-1, MNN-AE-2, and CNN, their abundances are the same as the reference ones, and ISTA uses the reference endmembers. The abundance maps obtained by all four methods are shown in Fig. 7, from which we can find that all of them get the similar abundance maps that are very close to the reference one.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

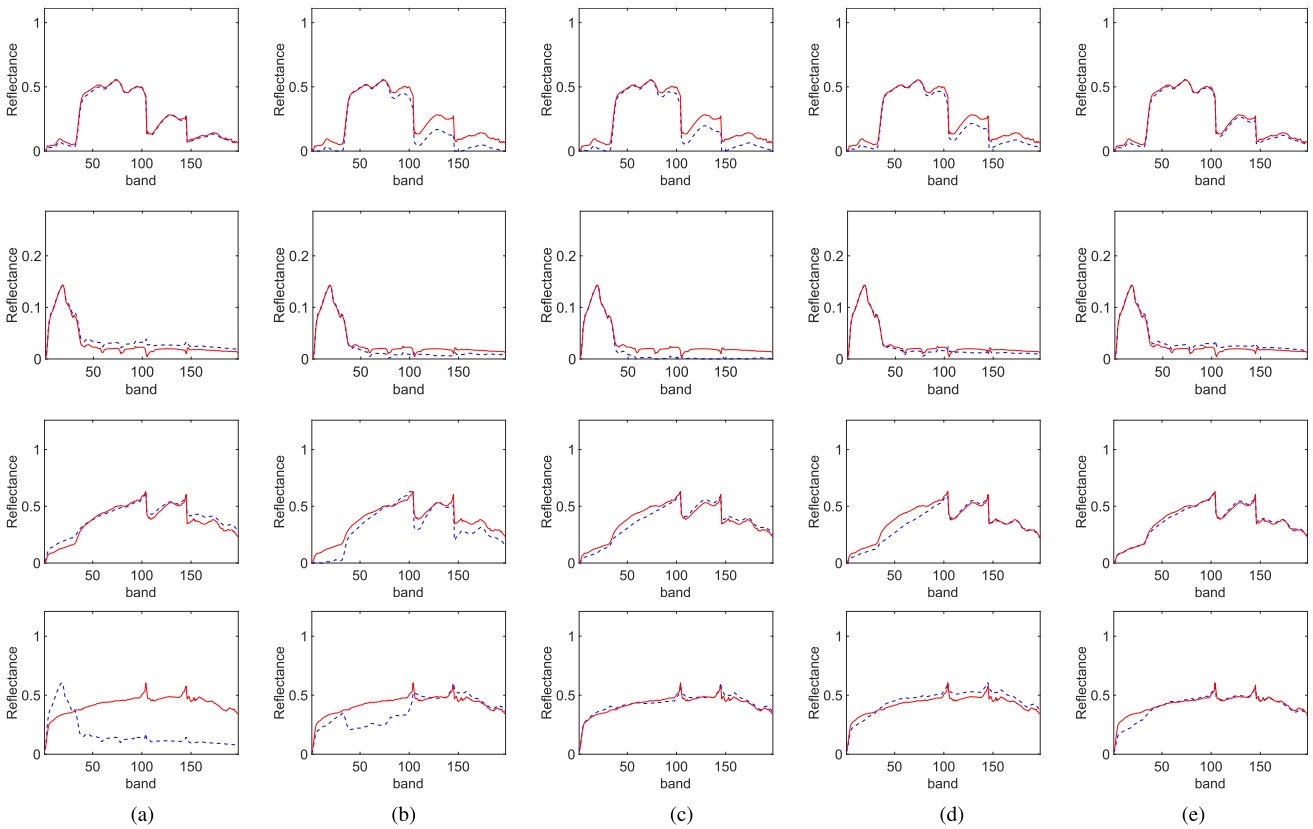


Fig. 12. Estimated endmembers by all five blind unmixing methods on the Jasper Ridge data set. (From top to bottom) Rows are the spectral signatures of tree, water, soil, and road, respectively. Solid lines: reference endmembers. Dashed lines: estimated endmembers. (a) R-NMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU-2.

b) HYDICE Urban Data Set: This HSI was collected by Hyperspectral Digital Image Collection Experiment (HYDICE) sensor in an urban area at Copperas Cove, TX, USA. It contains 307×307 pixels and 210 spectral bands ranging from 400 to 2500 nm with a spectral resolution of 10 nm. Before the experiment, noisy and water-absorption bands are removed, thus retaining a total of 162 spectral channels. It is assumed that there are four endmembers in this HSI, including tree, roof, asphalt, and grass. Its 80th band image is shown in Fig. 6(b). We also randomly select 1000 pixels for training. Fig. 8 demonstrates the results of all the methods, from which we can find that MNN-AE-1 and MNN-AE-2 give better visual results than CNN.

C. Blind Unmixing

In this section, we present the experimental results on synthetic and real-world data to evaluate the effectiveness of the proposed MNN-BU-1 and MNN-BU-2 for blind unmixing. We selected region-based NMF (R-NMF) [9], matrix-vector NTF (MV-NTF) [12], DAEN [33], and uDAS [34] as alternative methods for comparison. The architectures and parameters of all the competitors are set as recommended in the original articles. R-NMF enforces spectral–spatial structure into NMF to improve the unmixing performance. MV-NTF takes advantages of tensor in spectral–spatial structure preserving. Both DAEN and uDAS are based on autoencoder to generate endmembers and abundances. All the uDAS, DAEN, MNN-BU-1, and MNN-BU-2 have encoder–decoder structure, in which encoder is mapping the original spectral signature of the input pixel into its abundance vector and decoder remapping the abundance vector into its spectral signature. In our MNN-BU-1 and MNN-BU-2, the encoder is directly derived by unfolding ISTA algorithm, while both DAEN and uDAS are derived from the general autoencoder architecture. Therefore, our architectures have clearer interpretability and a stronger connection with LMM and ISTA than DAEN and uDAS.

1) Blind Unmixing on Synthetic Data: In this experiment, we analyze how the number of layers and the size of training data set influence the unmixing performance of our models, and compare our networks with other methods at different noise levels.

a) Impact of the number of layers: The experiments were carried out on the synthetic HSI generated with $\beta = 0.8$ and SNR $= 25$ dB. The 1000 training pixels are randomly selected, and the number of layers L is set from 1 to 9. Fig. 9 shows SAD and RMSE of MNN-BU-1 and MNN-BU-2 with respect to the number of layers. MNN-BU-2 is more stable than MNN-BU-1 when the number of layers changes, but both of them have good performance when the number of layers is small, thus implying the strong approximation capability of our networks even with very little size. In the following experiments, we set $L = 2$ for our blind unmixing networks.

b) Impact of the number of training samples: The synthetic data in this experiment are generated by setting $\beta = 0.8$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

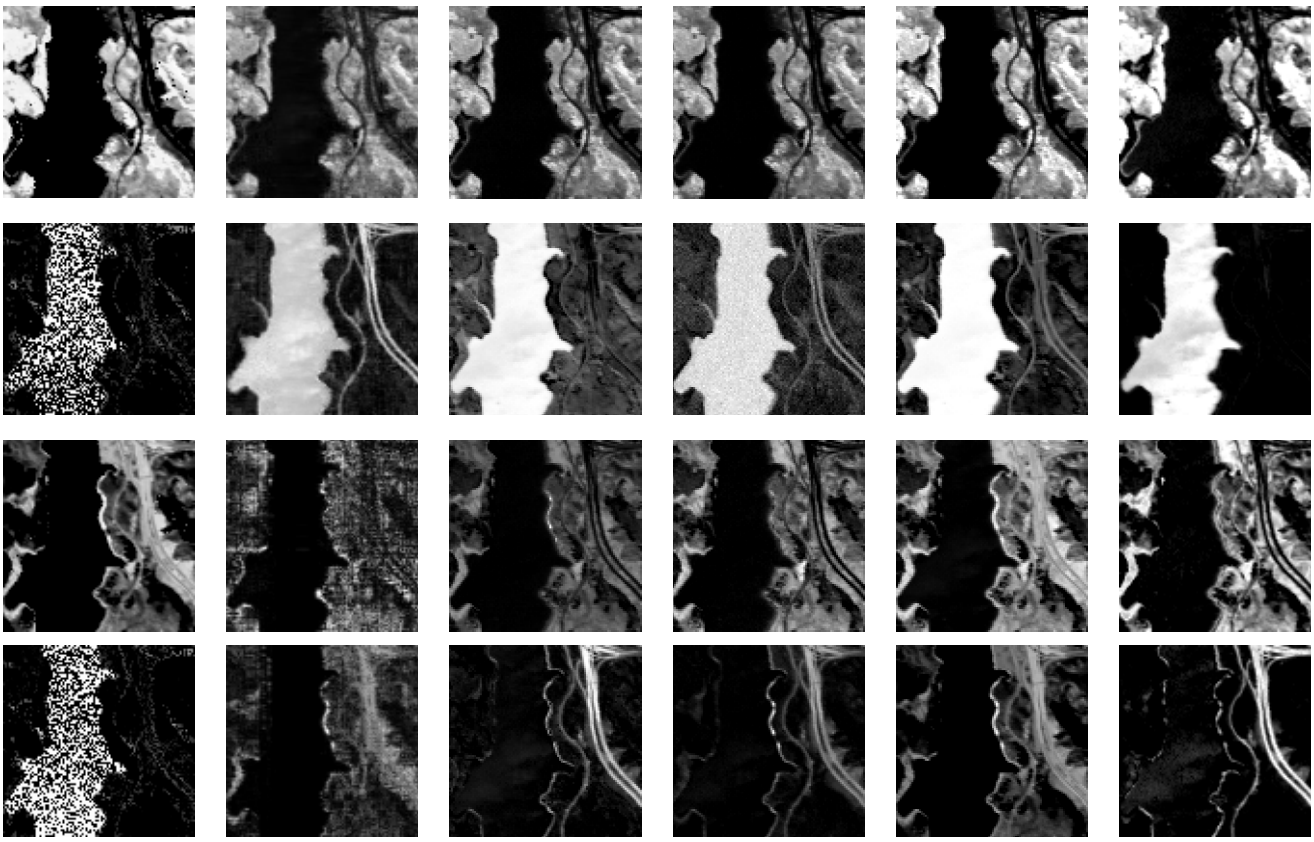


Fig. 13. Estimated abundance maps by all five blind unmixing methods on the Jasper Ridge data set. (From top to bottom) Rows are the abundance maps of tree, water, soil, and road, respectively. (a) R-NMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU-2. (f) Reference.

and SNR $=$ 25 dB. The training samples with the size $N = \{256, 500, 1000, 2000, 4096\}$ are selected from the data set, respectively, and all pixels are used for testing. The unmixing results of our networks are shown in Fig. 10. As the number of training samples increases, both MNN-BU-1 and MNN-BU-2 demonstrate better unmixing performance, especially SAD gets more rewards than RMSE as the size of training set increases, maybe due to the reason that endmember extraction fully lies on all training samples, while RMSE depends more on each individual pixel. As MNN-BU-2 is a little better than MNN-BU-1, we only use MNN-BU-2 in the following experiments.

c) Robustness to noise: This experiment aims to test the performance of all methods in noisy HSIs. We set $\beta = 0.8$ to produce the clean HSI, which is then contaminated by different levels of noises with SNR $= \{15$ dB, 20 dB, 25 dB, 30 dB, $\infty\}$. All 4096 pixels are used for training, which is the same as the experimental setting of DAEN and uDAS in their original articles. Fig. 11 shows the unmixing performance of all the compared methods, from which we find that MNN-BU-2 has the best performance among them. Specifically, MNN-BU-2 has a similar RMSE as DAEN but gives better SAD. Overall, NN methods including MNN-BU-2, DAEN, and uDAS provide noticeably better results than matrix and tensor factorization-based methods.

d) Running time comparison: The synthetic image is generated by setting $z = 8$, $\beta = 0.8$, and SNR $=$ 25 dB.

TABLE II
RUNNING TIME COMPARISON

	DAEN	R-NMF	MV-NTF	uDAS	MNN-BU-2
Time (s)	6276.4	43.6	14.4	3.1	9.3

We use all 4096 training samples for training. Table II gives the running time of all approaches. DAEN takes the highest time to converge because it has two autoencoders, respectively, used for initialization and blind source separation. Overall, MNN-BU-2 and uDAS require less time.

2) Blind Unmixing on Real-World Data: In the experiments, we randomly select 1000 pixels from an HSI to learn the parameters of MNN-BU-2, DAEN, and uDAS, use all pixels for testing, and run each experiment five times with different training sample selections. The experiments on synthetic data have demonstrated that as the number of training samples increases, the proposed method demonstrates better unmixing performance. However, more samples used for network training will cause more training time. As two real-world HSIs have more pixels than synthetic data set, we selected 1000 pixels in each HSI as training samples in order to balance the unmixing accuracy and running time. This setting is more applicable to real scenarios.

a) Jasper Ridge Data Set: Table III illustrates the SAD results of all methods. MNN-BU-2, DAEN, and uDAS provide more favorable performance than R-NMF and MV-NTF, which

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

TABLE III

MEANS AND STANDARD DEVIATIONS OF THE SAD ON JASPER RIDGE DATA SET

	R-NMF	MV-NTF	DAEN	uDAS	MNN-BU-2
Tree	**0.0407±0.80%**	0.2209 ± 2.17%	0.1774±0.82%	0.1511±1.37%	0.0468±0.32%
Water	0.1596±0.69%	0.2352±1.65%	0.3237±1.42	0.1742% ±11.39%	**0.1179±0.23%**
Soil	0.0867±1.12%	0.1752±4.67%	0.1123±2.03%	0.1639±6.09%	**0.0357±0.43%**
Road	0.7402±0.28%	0.1741±4.35%	**0.0588±1.59%**	0.0597±0.27%	0.0901±0.37%
Mean	0.2568±0.27%	0.2015±1.77%	0.1680± 0.43%	0.1373± 3.27%	**0.0726± 0.11%**

TABLE IV

MEANS AND STANDARD DEVIATIONS OF THE SAD ON URBAN DATA

	R-NMF	MV-NTF	DAEN	uDAS	MNN-BU-2
Asphalt	**0.0875±1.73%**	0.1720 ± 0.75%	1.0224±0.27%	0.2174±0.04%	0.1227±1.24 %
Grass	0.2129±7.91%	0.1974±2.97%	0.5122 ±0.29%	0.6862± 33.88%	**0.1234±0.88%**
Tree	0.1456±1.08%	0.1284±0.32%	**0.0801±0.01%**	0.0955±1.12 %	0.1113 ±0.20%
Roof	0.3300±3.01%	0.3608±3.66%	**0.1069±0.03%**	0.2037±2.37%	0.3558±3.80%
Mean	0.1940±1.11%	0.2147±0.96%	0.4304±0.02%	0.3007±8.79%	**0.1783±1.22%**

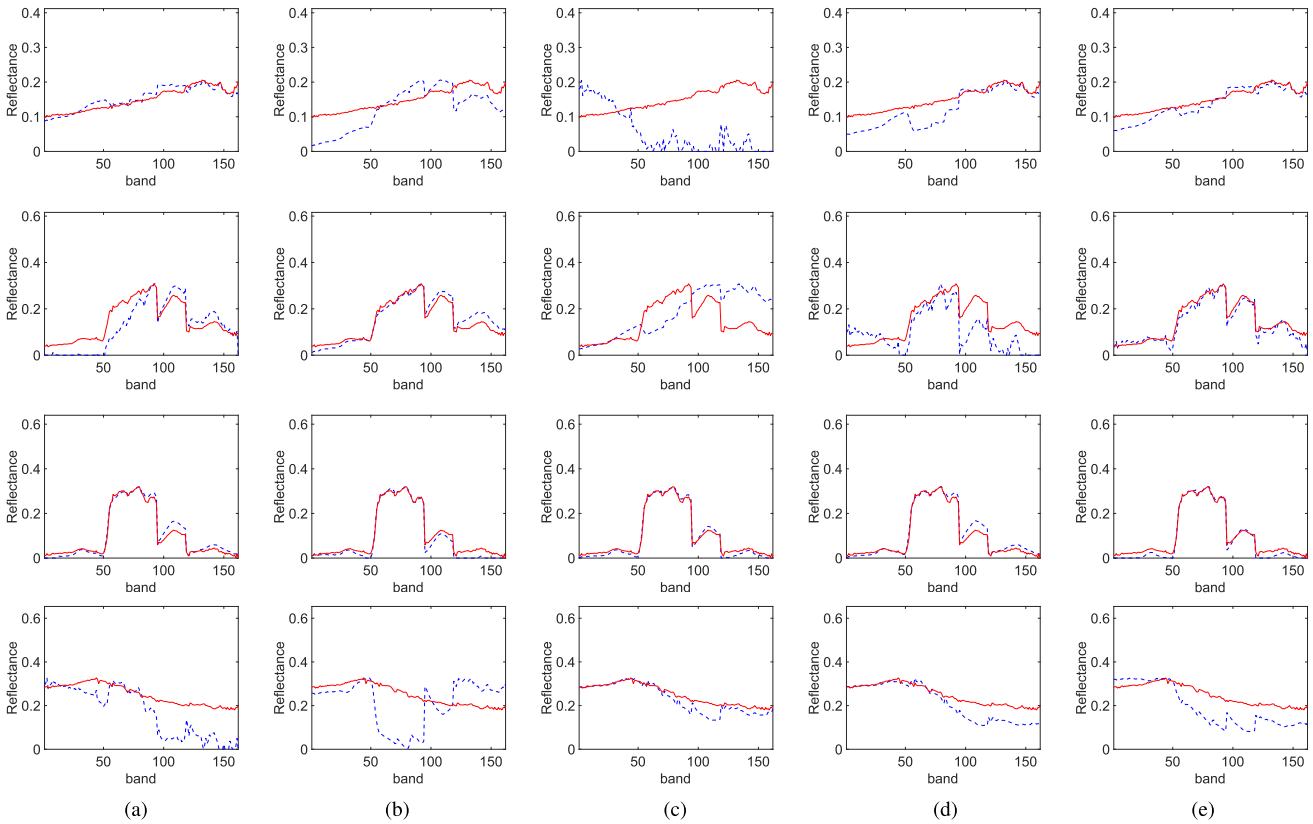


Fig. 14. Estimated endmembers by all five blind unmixing methods on HYDICE Urban data set. (From top to bottom) Rows are the spectral signatures of asphalt, grass, tree, and roof, respectively. Solid lines: reference endmembers. Dashed lines: estimated endmembers. (a) R-NMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU-2.

further demonstrates that NN-based blind unmixing methods have a distinct advantage over traditional unmixing techniques. Thanks to the network architecture inspired by LMM and ISTA, MNN-BU-2 achieves the best result among all methods. From Fig. 12 that shows the estimated and reference endmembers, it can be obviously found that the endmembers estimated by MNN-BU-2 are closest to reference ones. The abundance maps corresponding to all four endmembers are shown in Fig. 13.

b) HYDICE Urban Data Set: The comparison of all the methods in terms of SAD is shown in Table IV and Fig. 14. In this HSI data set, DAEN and uDAS do not outperform R-NMF and MV-NTF in terms of average SAD, but MNN-BU-2 still gives the best results among all methods. In particular, MNN-BU-2 is much better than DAEN for asphalt and grass and worse than DAEN for tree and roof, but our average performance is better than DAEN. Fig. 15 demonstrates the estimated abundance maps of all methods.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

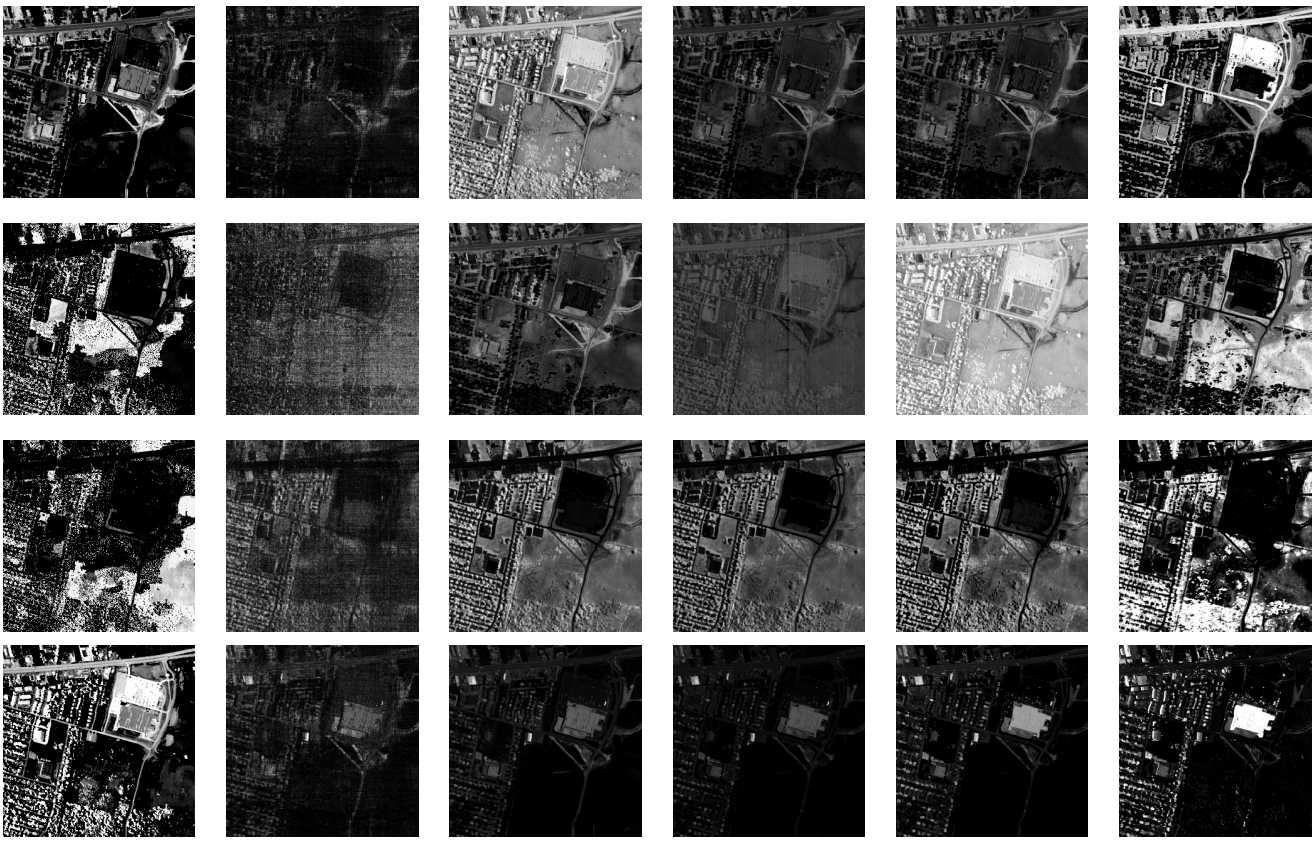


Fig. 15. Estimated abundance maps by all five blind unmixing methods on HYDICE Urban data set. (From top to bottom) Rows are the abundance maps of asphalt, grass, tree, and roof, respectively. (a) R-NMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU-2. (f) Reference.

V. Conclusion

This article discussed model-inspired NNs for spectral unmixing, which are based on LMM and ISTA. The first proposed network is directly unfolding ISTA of linear sparse coding, which learns an inference function to estimate the abundances of mixed pixels. Since there is no closed-form solution for linear sparse coding with nonnegativity constraint, iterative methods are the best choice and maybe the only choice in the past, but their computational costs are very expensive. Deep unfolding provides a novel technique to replace the iterations by a deep network architecture whose parameters are learned with a supervised data set. Compared with the iterative algorithm, this unfolded ISTA network has two superior properties: speed is very fast because no iteration is needed, and the performance can be improved by learning. We then developed an unsupervised version for blind unmixing by adding an endmember term as the last layer. In contrast to other NN-based blind unmixing approaches with autoencoder networks as the representative, its architecture is simpler and more interpretable as well. The advantages of the proposed methods have been demonstrated by the experiments on both synthetic and real hyperspectral data.

Model-inspired NNs applied for spectral unmixing is a new topic and will motivate more research. As our work in this article is an original and pioneering attempt, it provides plenty of room for future work in the following directions: 1) adopting other sparse coding algorithms such as AMP [40] and Fast ISTA [6] instead of ISTA to construct unfolded architectures; 2) using patch-based processing scheme or convolutional sparse coding model instead of pixel-based network architecture to incorporate spatial structure in HSIs into the networks; and 3) unfolding kernel-based sparse coding algorithms to extend the proposed methods to nonlinear mixture model.

References

[1] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[2] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[3] M.-D. Iordache, J. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.

[4] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[5] I. Daubechies, M. Defrise, and C. De Mol, "An iterative thresholding algorithm for linear inverse problems with a sparsity constraint," *Commun. Pure Appl. Math.*, vol. 57, no. 11, pp. 1413–1457, Nov. 2004.

[6] A. Beck and M. Teboulle, "A fast iterative shrinkage-thresholding algorithm for linear inverse problems," *SIAM J. Imag. Sci.*, vol. 2, no. 1, pp. 183–202, Jan. 2009.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.

[7] S. Jia and Y. Qian, "Constrained nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 1, pp. 161–173, Jan. 2009.

[8] J. Sigurdsson, M. O. Ulfarsson, and J. R. Sveinsson, "Blind hyperspectral unmixing using total variation and ℓ_q sparse regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6371–6384, Nov. 2016.

[9] L. Tong, J. Zhou, X. Li, Y. Qian, and Y. Gao, "Region-based structure preserving nonnegative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1575–1588, Apr. 2017.

[10] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.

[11] R. Huang, X. Li, and L. Zhao, "Spectral–spatial robust nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8235–8254, Oct. 2019.

[12] Y. Qian, F. Xiong, S. Zeng, J. Zhou, and Y. Y. Tang, "Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1776–1792, Mar. 2017.

[13] F. Xiong, Y. Qian, J. Zhou, and Y. Y. Tang, "Hyperspectral unmixing via total variation regularized nonnegative tensor factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2341–2357, Apr. 2019.

[14] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.

[15] W. Wang and Y. Qian, "Adaptive $L_{1/2}$ sparsity-constrained NMF with half-thresholding algorithm for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2618–2631, Jun. 2015.

[16] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[17] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[18] J. Chen, S. Chen, P. Zhou, and Y. Qian, "Deep neural network based hyperspectral pixel classification with factorized spectral–spatial feature representation," *IEEE Access*, vol. 7, pp. 81407–81418, Jun. 2019.

[19] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[20] X. Zhang, Y. Sun, J. Zhang, P. Wu, and L. Jiao, "Hyperspectral unmixing via deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1755–1759, Nov. 2018.

[21] S. Ozkan, B. Kaya, and G. B. Akar, "EndNet: Sparse AutoEncoder network for endmember extraction and hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 482–496, Jan. 2019.

[22] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, May 2015.

[23] J. R. Hershey, J. Le Roux, and F. Weninger, "Deep unfolding: Model-based inspiration of novel deep architectures," Sep. 2014, *arXiv:1409.2574*. [Online]. Available: http://arxiv.org/abs/1409.2574

[24] S. Wisdom, T. Powers, J. Pitton, and L. Atlas, "Building recurrent networks by unfolding iterative thresholding for sequential sparse recovery," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, New Orleans, LA, USA, Mar. 2017, pp. 4346–4350.

[25] P. Gong, C. Zhang, Z. Lu, J. Huang, and J. Ye, "A general iterative shrinkage and thresholding algorithm for non-convex regularized optimization problems," in *Proc. 30th Int. Conf. Mach. Learn. (ICML)*, Atlanta, GA, USA, 2013, pp. 37–45.

[26] Q. Qian, F. Xiong, and J. Zhou, "Deep unfolded iterative shrinkage-thresholdingmodel for hyperspectral unmixing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Yokohama, Japan, Jul. 2019, pp. 2151–2154.

[27] H. Lee, A. Battle, R. Raina, and A. Y. Ng, "Efficient sparse coding algorithms," in *Proc. Adv. Neural Inf. Process. Syst.*, 2006, pp. 801–808.

[28] S. Foucart, "Learning optimal nonlinearities for iterative thresholding algorithms," *SIAM J. Numer. Anal.*, vol. 49, no. 6, pp. 2543–2563, 2011.

[29] G. A. Licciardi and F. Del Frate, "Pixel unmixing in hyperspectral data by means of neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4163–4172, Nov. 2011.

[30] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.

[31] M. Wang, M. Zhao, J. Chen, and S. Rahardja, "Nonlinear unmixing of hyperspectral data via deep autoencoder networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1467–1471, Sep. 2019.

[32] Y. Su, A. Marinoni, J. Li, J. Plaza, and P. Gamba, "Stacked nonnegative sparse autoencoders for robust hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1427–1431, Sep. 2018.

[33] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.

[34] Y. Qu and H. Qi, "UDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.

[35] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proc. 27th Int. Conf. Mach. Learn. (ICML)*, Haifa, Israel, 2010, pp. 399–406.

[36] J. L. Roux, J. R. Hershey, and F. J. Weninger, "Deep NMF for speech separation," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, Brisbane, QLD, Australia, 2015, pp. 66–70.

[37] S. Wisdom, J. Hershey, J. Le Roux, and S. Watanabe, "Deep unfolding for multichannel source separation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Shanghai, China, Mar. 2016, pp. 121–125.

[38] R. Vemulapalli, O. Tuzel, and M.-Y. Liu, "Deep Gaussian conditional random field network: A model-based deep network for discriminative denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4801–4809.

[39] A. V. Goldberg, S. Hed, H. Kaplan, R. E. Tarjan, and R. F. Werneck, "Deep unfolded IRLS-ADMM network for classification and sparse feature selection," *Int. J. Mach. Learn. Comput.*, vol. 8, no. 3, pp. 241–251, Jun. 2018.

[40] M. Borgerding, P. Schniter, and S. Rangan, "AMP-inspired deep networks for sparse linear inverse problems," *IEEE Trans. Signal Process.*, vol. 65, no. 16, pp. 4293–4308, Aug. 2017.

[41] J.-T. Chien and C.-H. Lee, "Deep unfolding for topic models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 2, pp. 318–331, Feb. 2018.

[42] P. B. P. Sprechmann and G. Sapiro, "Learning efficient structured sparse models," in *Proc. 29th Int. Conf. Mach. Learn. (ICML)*, Edinburgh, U.K., 2012, pp. 615–622.

[43] J. Sun, Y. Yang, H. Li, and Z. Xu, "Deep ADMM-Net for compressive sensing MRI," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 10–18.

[44] S. Mandt, M. D. Hoffman, and D. M. Blei, "Stochastic gradient descent as approximate Bayesian inference," *J. Mach. Learn. Res.*, vol. 18, pp. 1–35, Jan. 2017.

[45] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Nov. 2011.

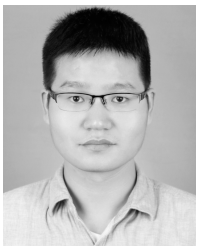


Yuntao Qian (Member, IEEE) received the B.E. and M.E. degrees in automatic control from Xi'an Jiaotong University, Xi'an, China, in 1989 and 1992, respectively, and the Ph.D. degree in signal processing from Xidian University, Xi'an, in 1996.

From 1996 to 1998, he was a Post-Doctoral Fellow with Northwestern Polytechnical University, Xi'an. Since 1998, he has been with the College of Computer Science, Zhejiang University, Hangzhou, China, where he became a Professor in 2002. From 1999 to 2001 and in 2006, 2010, 2013, 2015–2016, and 2018, he was a Visiting Professor with Concordia University, Montreal, QC, Canada, Hong Kong Baptist University, Hong Kong, Carnegie Mellon University, Pittsburgh, PA, USA, the Canberra Research Laboratory, NICTA, Canberra, ACT, Australia, Macau University, Taipa, Macau, and Griffith University, Nathan, QLD, Australia. His research interests include machine learning, signal and image processing, pattern recognition, and hyperspectral imaging.

Dr. Qian is an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.



Fengchao Xiong received the B.S. degree in software engineering from Shandong University, Jinan, China, in 2014. He is pursuing the Ph.D. degree with the College of Computer Science, Zhejiang University, Hangzhou, China.

His research interests include hyperspectral image processing, machine learning, and pattern recognition.



Qipeng Qian is pursuing the bachelor's degree with the School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai, China.

He is interested in mathematics in image processing and machine learning.



Jun Zhou (Senior Member, IEEE) received the B.S. degree in computer science and the B.E. degree in international business from the Nanjing University of Science and Technology, Nanjing, China, in 1996 and 1998, respectively, the M.S. degree in computer science from Concordia University, Montreal, QC, Canada, in 2002, and the Ph.D. degree from the University of Alberta, Edmonton, AB, Canada, in 2006.

He was a Research Fellow with the Research School of Computer Science, The Australian National University, Canberra, ACT, Australia, and a Researcher with the Canberra Research Laboratory, NICTA, Canberra. He is an Associate Professor with the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia. His research interests include pattern recognition, computer vision, and spectral imaging with their applications to remote sensing and environmental informatics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:42:44 UTC from IEEE Xplore. Restrictions apply.